IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 7: Edge Detection Techniques

Complete the table:

Sobel – Type: First-order derivative | Noise Sensitivity: Moderate

Prewitt – Type: First-order derivative | Noise Sensitivity: Moderate

Canny – Type: Multi-stage edge detector | Noise Sensitivity: Low

Roberts – Type: First-order derivative | Noise Sensitivity: High

Why is Canny called an optimal edge detector?

Canny is called an **optimal edge detector** because it is designed to detect edges accurately while reducing noise and avoiding unnecessary or false edges. It uses a **multi-stage process** that includes Gaussian smoothing, gradient calculation, non-maximum suppression, double thresholding, and edge tracking by hysteresis.

The Gaussian filter first reduces noise, while gradient calculation identifies the strength and direction of edges. **Non-maximum suppression** makes the detected edges thin and precise. Double thresholding classifies edges as strong or weak, and edge tracking by hysteresis connects weak edges that are related to strong edges. This helps preserve important edges while removing unwanted noise and false responses.

Therefore, Canny provides a good balance of **accurate edge detection, good localization, and noise reduction**, making it one of the most effective and widely used edge detection techniques.